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标题	Mechanochemical Copper-Catalyzed Azide – Alkyne Cycloaddition (CuAAC)
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摘要（原文节选）

Over the last 15 years, significant progress has been made in developing mechanochemical approaches for the copper-catalyzed azide – alkyne cycloaddition (CuAAC) reaction, utilizing various starting materials and different copper sources and process parameters. This synopsis systematically discusses developments in the mechanochemical CuAAC reaction, aiming not only to highlight the benefits of employing this approach for designing molecules featuring a 1,2,3-triazole motif but also to provide a comprehensive guide f ⋯

数据表 · 2（24 行，含数值行 17 行）

Summary of Key Information for the Developed Approaches of CuAAC Click-Mechanochemical Reactions

Figure in this synopsis	Starting material type - alkyne	The second starting material(s) type	Copper source	Molar equiv of copper source	Instrument to conduct a mecha nochemical process	Reaction time	Yield (%)	Ref
Figure	various benzene derivatives, various aliphatic derivatives	various aliphatic and aromatic azides	Cu(OAc)2	0.05	planetary ball mill (ZrO2 beaker and milling balls)	5 – 20 min	75 – 97	
Figure	various benzene derivatives, various aliphatic derivatives	benzyl bromide derivatives, sodium azide	copper reaction vessel with copper milling balls	N/A	ball mill	16 h	33 – 95	
Figure a	various benzene derivatives, various aliphatic derivatives	benzyl halide derivatives, secondary halides, sodium azide	Cu/Al2O3	0.01	ball mill	16 h	70 – 96	
Figure b	various benzene derivatives, various aliphatic derivatives	various boronic acid derivatives, sodium azide	Cu/Al2O3 ,	0.01	ball mill	2 h	83 – 91	
Figure a	ethynylbenzene, various aliphatic derivatives	octyl azide, various aliphatic azides	Cu powder	1.00	planetary ball mill	5 – 10 min	88 – 99	
Figure b	ethynylbenzene	β -cyclodextrin monoazide	Cu powder	0.1	planetary ball mill	30 min	81	
Figure	mono-N-proparg ylamide substituted azobenzenes	5 ′ -azido-5 ′ -de oxythymidine	copper milling balls	N/A	HSVBM (LAG, ethyl acetate)	18 h	63 – 80 ,	
Figure a	propargylated sugars	hexadecyl azide	CuSO4 · 5H2O	0.2	planetary ball mill	2 – 6 h	74 – 89	
Figure b	propargylated sugars	various aliphatic azides	CuSO4 · 5H2O	0.2	planetary ball mill	N/A	85 – 92	
Figure	propargylated quinoline derivatives	phenyl azide, p-halogenated benzyl azide	Cu(OAc)2 · H2 O	0.05	ball mill	3.5 h	57 – 77	
CuI	0.02	79 – 92						
brass balls	N/A	76 – 97						
Figure	dendritic-like triazine derivative	sulfonated, aliphatic azides	presynthesized copper catalyst (see structure in Figure)	0.06	not specified (LAG, water)	1 h	94 – 99	
Figure	various benzene derivatives, various aliphatic derivatives	azide derivative of penicillin	CuSO4	0.2	planetary ball mill	3 h	57 – 85	
Figure	propargyl-contain ing oxindole derivatives	benzyl azide or its derivatives	CuO nanoparticles	0.05	ball mill	30 min	87 – 92	

Figure in this synopsis	Starting material type - alkyne	The second starting material(s) type	Copper source	Molar equiv of copper source	Instrument to conduct a mechanochemical process	Reaction time	Yield (%)	Ref
Figure a	aromatic or heterocyclic alkynes	derivatives bearing an α-bromo-carbonyl motif	copper beads	N/A	HSVBM	2 – 3 h	44 – 93	
Figure b	aromatic or heterocyclic alkynes	p-toluenesulfonyl azide	copper beads	N/A	HSVBM	1.5 – 3 h	63 – 83	
Figure c	ethyl propiolate	derivatives of Rufinamide bearing an azide group	copper beads	N/A	HSVBM	2 h	67 – 79	
Figure	respective propargylated resorcine derivatives (dendritic)	azide-derivative of glucuronic acid methyl ester	CuSO4 · 5H2O	0.2	planetary ball mill	11 h	65 – 70	
Figure	various phenyl-acetylene derivatives, 2-ethynyl-thiophene	benzyl bromide derivatives, sodium azide, diaryl diselenides	CuI	0.1	HSVBM	15 min	42 – 87	
Figure	propargylated quinoline/quinolone derivatives	1,1′-disubstituted ferrocene azides	Cu(OAc)2	0.05	ball mill (LAG, methanol)	30 min	59 – 93	
Figure	aromatic or aliphatic alkynes	benzyl bromide, sodium azide	presynthesized copper catalyst (see structure in Figure)	0.015	ball mill	3 h	19 – 78	
Figure	propargylated GO	PEG derivative bearing maleimide and azide end-groups	CuSO4	0.06 (with respect to the azide)	mixer mill	1 h	-	
Figure	natural lignin propargyl derivative	aliphatic or aromatic azides	Cu(OAc)2	0.08	planetary ball mill (ZrO2 milling balls)	30 min	37 – 77	

脚注（逐字）：aInert milling auxiliary = SiO2 (quartz sand), sodium ascorbate used in selected experiments.bEstimated by1H NMR.cPrepared by stirring the mixture of CuSO4 · 5H2O with basic alumina, followed by evaporation and drying.dIsolated yields.ePotassium carbonate (K2CO3) added in the first reaction step.fHigh

正文对应句（逐字）：Regarding further studies on searching fundamental conditions and reaction set-ups for CuAAC click-mechanochemistry, in 2013, two reports on the multicomponent approach toward CuAAC were published independently by the groups of Mack and Ranu. In these studies, except the alkyne starting material, be

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